

Acceleration of high-energy particles (First and second order Fermi acceleration) (Lecture 5)

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Outline of this lecture

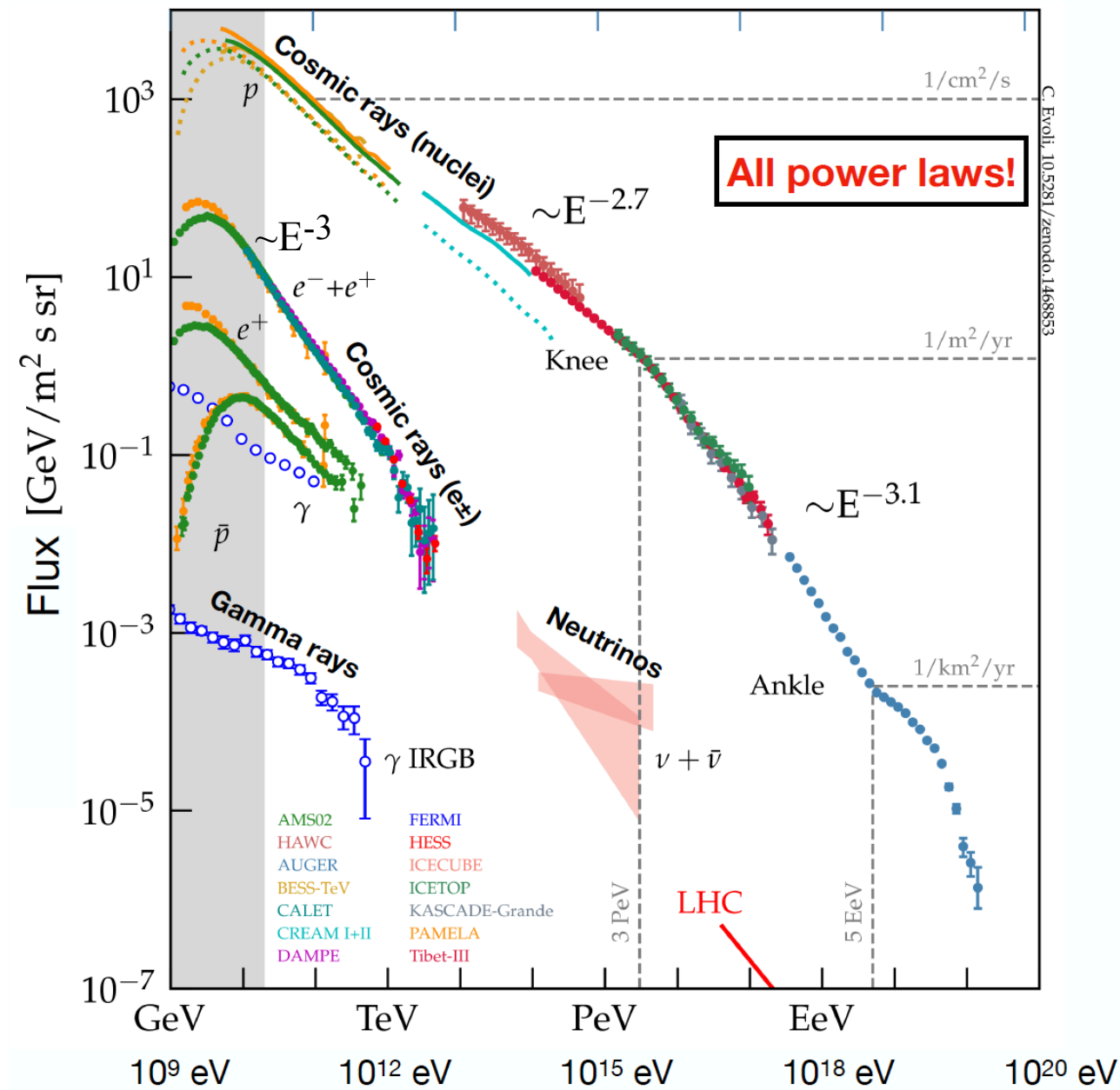
- Introduction
- Second order Fermi acceleration
- First order Fermi acceleration
- Dynamics of charged particles in magnetic field
- Shock waves
- Diffusive shock acceleration

Acceleration of high-energy particles

Introduction

Process of acceleration of high-energy particles must account for these features:

- (1) Energy spectra of cosmic-ray nuclei and electrons follow power-law behavior $dN(E) \propto E^{-x} dE$
- (2) Acceleration of cosmic-ray nuclei upto $\sim 10^{20}$ eV energies.

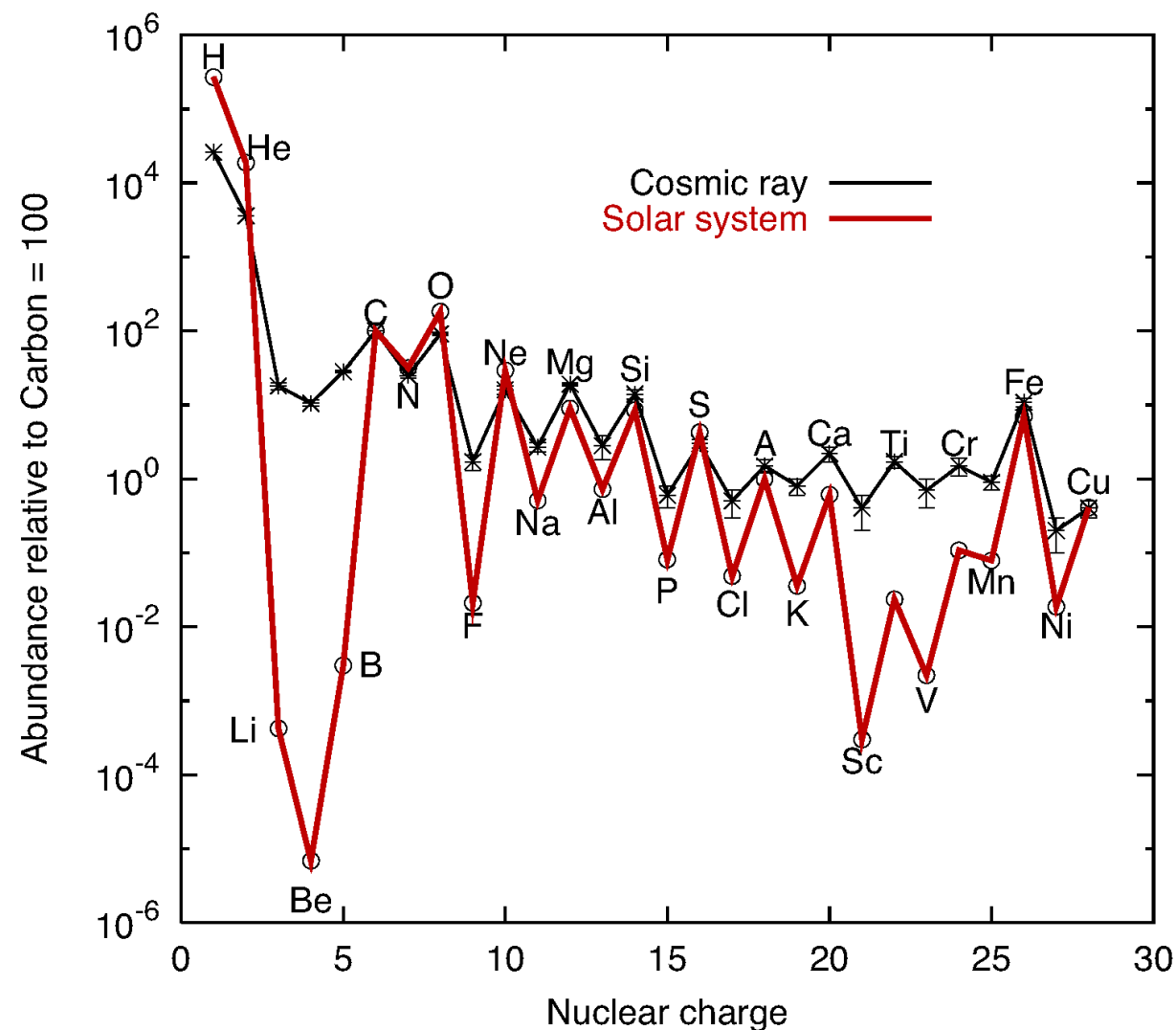


Acceleration of high-energy particles

Introduction

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- (1) Energy spectra of cosmic-ray nuclei and electrons follow power-law behavior $dN(E) \propto E^{-x} dE$
- (2) Acceleration of cosmic-ray nuclei unto $\sim 10^{20}$ eV energies.
- (3) Chemical abundances of cosmic rays follow similar to the cosmic (solar) abundances of elements.



Acceleration of high-energy particles

Introduction

Acceleration mechanisms:

- Dynamic: Acceleration by collisions.
- Electromagnetic: Acceleration by electric fields.
-

Often, there is no clear distinction with these as all processes involve charge particles which are closely tied to magnetic field lines.

General expression for acceleration of a charged particle in electric and magnetic fields:

$$\frac{d}{dt}(\gamma m \mathbf{v}) = e(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

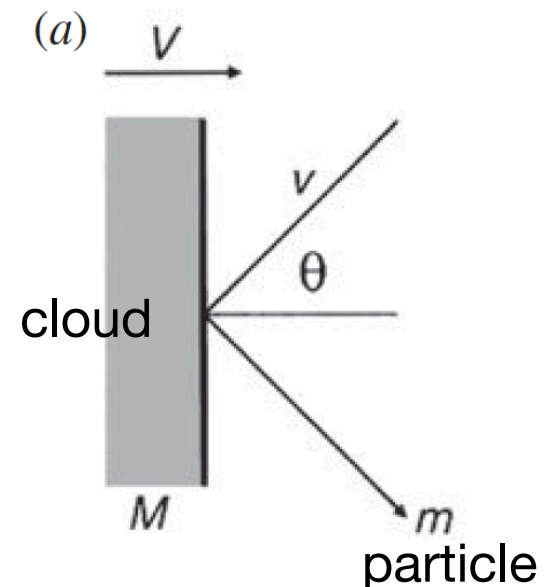
In most astrophysical environments, time varying magnetic fields $\text{curl } \mathbf{E} = -\partial \mathbf{B} / \partial t$ induce electric field which can accelerate particle. But, it is not clear if this mechanism can accelerate particles to the very high energies that has been observed.

Acceleration of high-energy particles

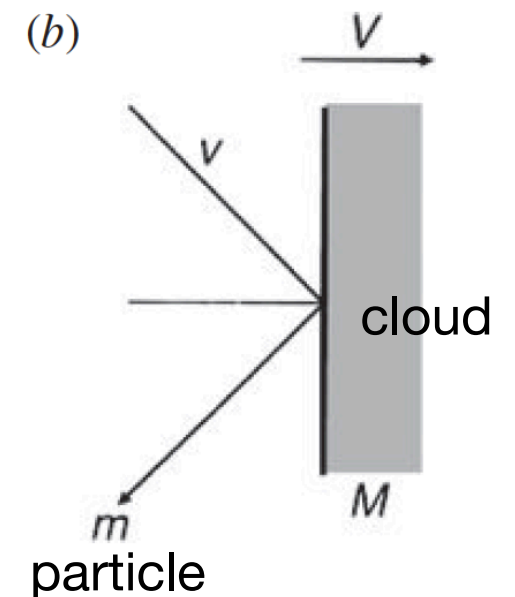
Second order Fermi acceleration

- First proposed by Enrico Fermi in 1949.
- Particle acceleration to high energies through scattering by interstellar clouds.
- Charged particles are reflected by “magnetic mirrors” (interstellar clouds).
- The mirrors move randomly with some velocities.
- During head-on collision (elastic), particle gain energy.
- Particle lost energy during trailing collision.
- Probability of head-on collision is slightly greater than trailing collision.
- Fermi showed that particles gained energy overall in these reflections.
- If the particles remain confined within the acceleration region for some time, a power-law distribution of particle energies is obtained.

Head-on collision



Trailing collision



Acceleration of high-energy particles

Second order Fermi acceleration

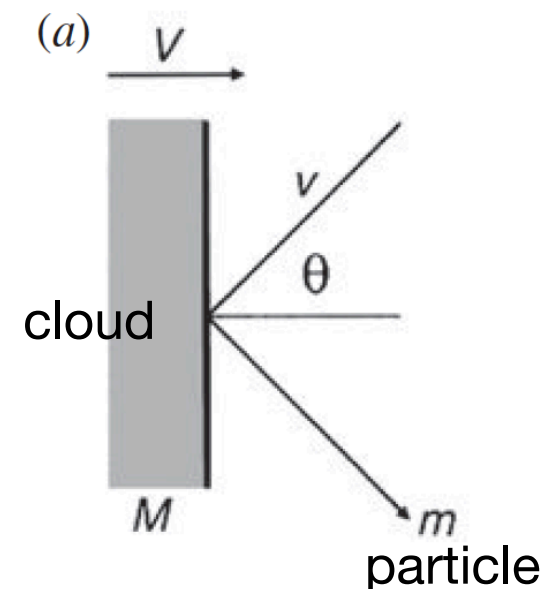
In the reference frame of a cloud moving with velocity V , the energy of the particle is (using Lorentz transformation):

$$E' = \gamma_V (E + V p \cos \theta), \quad \text{where} \quad \gamma_V = \left(1 - \frac{V^2}{c^2}\right)^{-1/2}$$

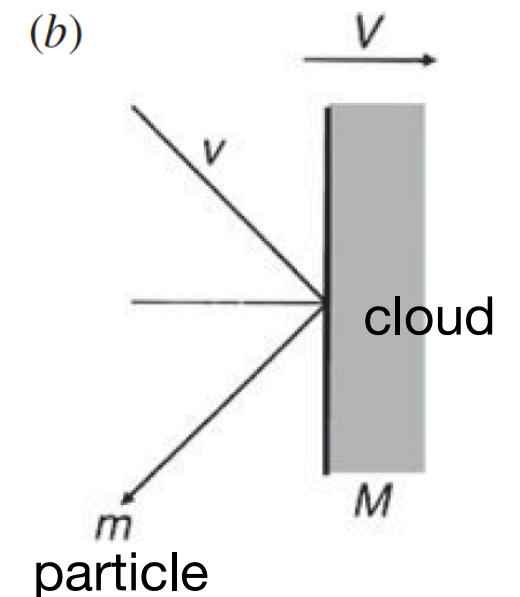
And the momentum along the x-axis:

$$p'_x = p' \cos \theta' = \gamma_V \left(p \cos \theta + \frac{V E}{c^2} \right)$$

Head-on collision



Trailing collision



Acceleration of high-energy particles

Second order Fermi acceleration

In the reference frame of a cloud moving with velocity V , the energy of the particle is (using Lorentz transformation):

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And the momentum along the x-axis:

$$p'_x = p' \cos \theta' = \gamma_V \left(p \cos \theta + \frac{V E}{c^2} \right)$$

Assuming the cloud is very massive, its velocity is unchanged in the collision. So, the particle's energy does not change in the cloud's frame. Only, its momentum in the x-direction is reversed.

$$E'_{\text{before}} = E'_{\text{after}}, \quad p'_x \rightarrow -p'_x.$$

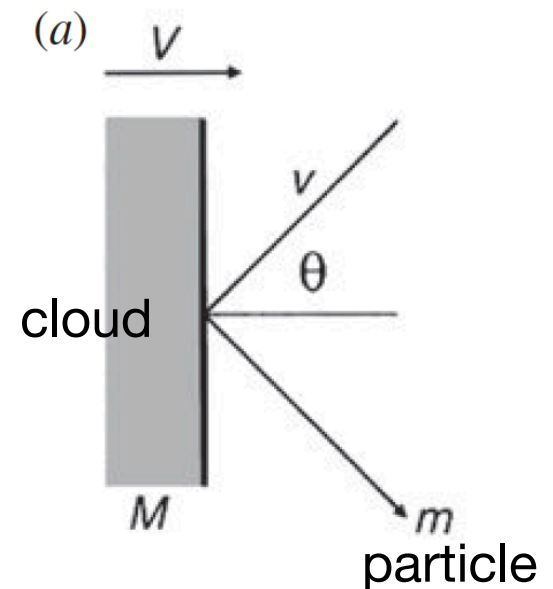
Transforming back to the lab frame, the particle energy becomes:

$$E'' = \gamma_V (E' + V p'_x)$$

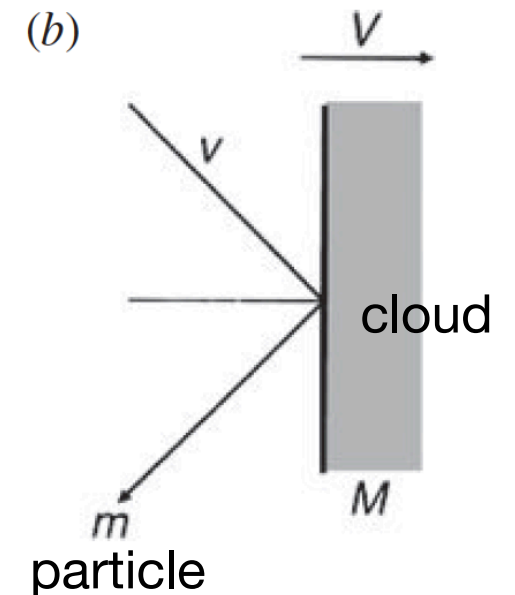
Substituting E' and p'_x and recalling $p_x/E = v \cos \theta/c^2$

$$\text{We get, } E'' = \gamma_V^2 E \left[1 + \frac{2V v \cos \theta}{c^2} + \left(\frac{V}{c}\right)^2 \right]$$

Head-on collision



Trailing collision



Acceleration of high-energy particles

Second order Fermi acceleration

For non-relativistic clouds, $\gamma_V \sim 1$.

Thus, from the previous page, $E'' = \gamma_V^2 E \left[1 + \frac{2Vv \cos \theta}{c^2} + \left(\frac{V}{c} \right)^2 \right]$

$$\begin{aligned} \text{We get, } E'' - E &= E \left[\frac{2Vv \cos \theta}{c^2} + 2 \left(\frac{V}{c} \right)^2 \right] \\ \Rightarrow \frac{\Delta E}{E} &= \left[\frac{2Vv \cos \theta}{c^2} + 2 \left(\frac{V}{c} \right)^2 \right] \end{aligned}$$

Acceleration of high-energy particles

Second order Fermi acceleration

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$$\Rightarrow \frac{\Delta E}{E} = \left[\frac{2Vv \cos \theta}{c^2} + 2 \left(\frac{V}{c} \right)^2 \right]$$

As we can see, the fractional change in energy can be both positive and negative. We need to check what is the net result. Basically, we need to integrate $\Delta E/E$ over all possible incident angles taking into account the probability of the collision at each angle, $P(\cos\theta) \sim k[1 + (V/c) \cos \theta]$

The condition $\int_{-1}^1 P(\mu) d\mu = 1$ gives $k = 1/2$, where $\mu = \cos\theta$.

Acceleration of high-energy particles

Second order Fermi acceleration

Finally, the average energy gain per collision is obtained by averaging over all angles from 0 to 180°,

$$\left\langle \frac{\Delta E}{E} \right\rangle = \int_{-1}^1 \left[\frac{2V\mu}{c} + 2 \left(\frac{V}{c} \right)^2 \right] P(\mu) d\mu$$

$$\Rightarrow \left\langle \frac{\Delta E}{E} \right\rangle = \frac{8}{3} \left(\frac{V}{c} \right)^2$$

- *In every particle-cloud interaction, the particle can either gain or lose energy.
- *Probability of head-on collision is slightly greater than trailing collision.
- *On average, the particle gain energy, but scales as second order in the cloud's velocity. So, the name “Second order Fermi acceleration”.

Acceleration of high-energy particles

Second order Fermi acceleration

Now, how about the shape of the particle spectrum produce by Fermi acceleration?

If the mean free path between clouds along a field line is L , the time between collisions is $L/(c \cos \phi)$ where ϕ is the pitch angle of the particle with respect to the magnetic field direction. Averaging $\cos \phi$ over the pitch angle ϕ , the average time between collisions is $2L/c$.

The average rate of energy increase is
$$\frac{dE}{dt} = \frac{4}{3} \left(\frac{V^2}{cL} \right) E = \alpha E$$

Acceleration of high-energy particles

Second order Fermi acceleration

Now, how about the shape of the particle spectrum produced by Fermi acceleration?

$$\text{Let's write } b(E) = \frac{dE}{dt} = \frac{4}{3} \left(\frac{V^2}{cL} \right) E = \alpha E$$

The particle is assumed to remain within the accelerating region for a time τ_{esc} . The resulting spectrum can be found from the gain-loss balance equation:

$$\begin{aligned} -\frac{d}{dE} [b(E)N(E)] - \frac{N(E)}{\tau_{\text{esc}}} &= 0 \\ \Rightarrow -\frac{d}{dE} [\alpha E N(E)] - \frac{N(E)}{\tau_{\text{esc}}} &= 0 \end{aligned}$$

Differentiating and rearranging this equation,
$$\frac{dN(E)}{dE} = - \left(1 + \frac{1}{\alpha \tau_{\text{esc}}} \right) \frac{N(E)}{E}$$

Solving, we get: $N(E) = \text{constant} \times E^{-x}$ where $x = 1 + (\alpha \tau_{\text{esc}})^{-1}$

The Fermi acceleration mechanism results in a power-law energy spectrum.

Acceleration of high-energy particles

First order Fermi acceleration

- *The process will be more efficient if there is only head-on collision.
- *Then, there will be only energy gain, no energy loss.
- *On average, the particle will gain energy as V/c . So, the name “First order Fermi acceleration”.

Acceleration of high-energy particles

First order Fermi acceleration

Recalling the particle energy in the rest frame of the scatterer moving with velocity V ,

$$E' = \gamma_V (E + V p \cos \theta), \quad \text{where} \quad \gamma_V = \left(1 - \frac{V^2}{c^2}\right)^{-1/2} \sim 1 \text{ because } V \ll c$$
$$\Rightarrow (E' - E)/E = \frac{V}{c} \cos \theta \quad E = pc, \text{ because particle is relativistic.}$$
$$\Rightarrow \frac{\Delta E}{E} = \frac{V}{c} \cos \theta \quad \Rightarrow p = E/c$$

Acceleration of high-energy particles

First order Fermi acceleration

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Now, the probability of collision for particle arriving at an angle θ is given by

$$p(\theta) = 2 \sin \theta \cos \theta \, d\theta$$

Finally, the average energy gain per collision is obtained by averaging over all angles,

$$\left\langle \frac{\Delta E}{E} \right\rangle = \frac{V}{c} \int_0^{\pi/2} 2 \cos^2 \theta \sin \theta \, d\theta = \frac{2}{3} \frac{V}{c}$$

Acceleration of high-energy particles

First order Fermi acceleration

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The energy gain scales as first order in V/c . So, the name “First order Fermi acceleration”.

This is the acceleration process that is believed to operate in most astrophysical sources, often referred to as “Diffusive shock acceleration”.

Acceleration of high-energy particles

First order Fermi acceleration

This is the mechanism of particle acceleration in most astrophysical environments associated with strong shock waves. The mechanism is often referred to as *diffusive shock acceleration*.

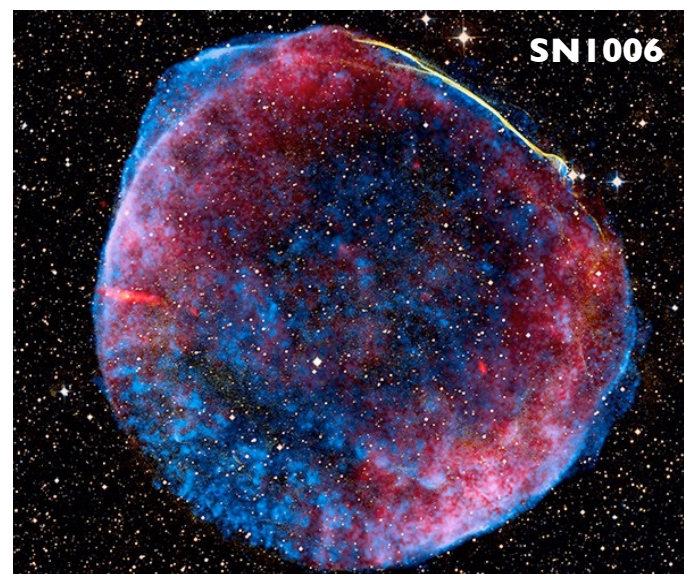
Developed by a number of researchers independently in the late 1970s:

Krymskii, G. F. 1977, Akad. Nauk SSSR Dokl., 234, 1306

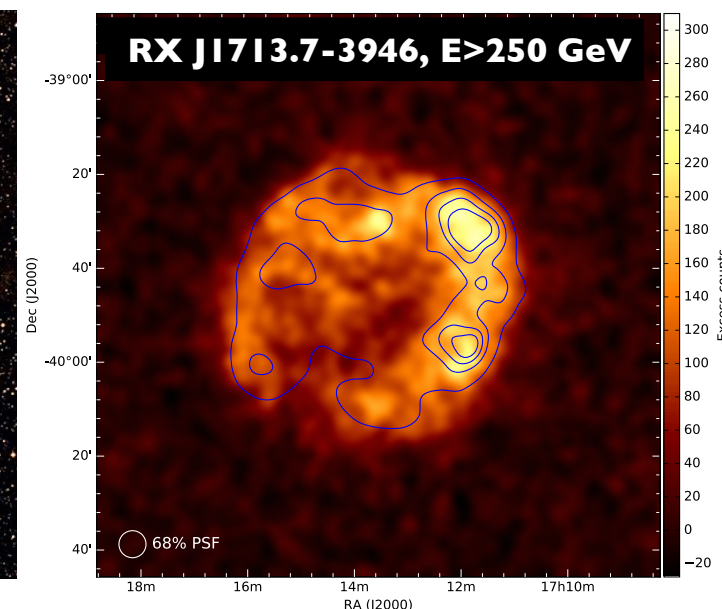
Bell, A. R. 1978, MNRAS 182, 147

Blandford, R., & Ostriker, J. P. 1978, ApJ, 221, L29

For example, the shock waves which propagate through the interstellar medium ahead of the supersonic shells of supernova remnants. Materials ejected in the supernova explosion produced tremendous shock waves that heated gas to millions of degrees and accelerated particles to extremely high energies.



Chandra's X-ray image



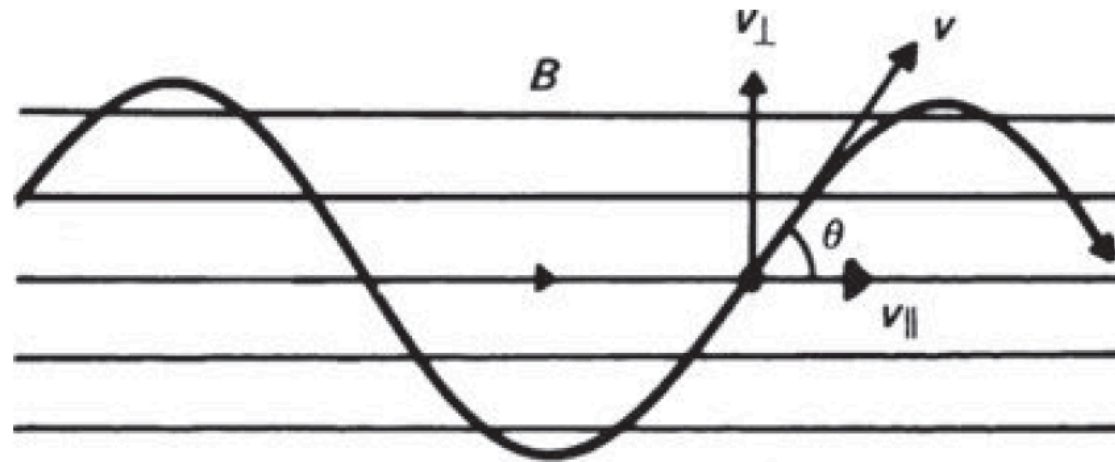
H.E.S.S. gamma-ray image

Acceleration of high-energy particles

Dynamics of charged particles in a uniform static magnetic field

Motion of a particle of rest mass m_0 , charge ze and velocity $\mathbf{v}$ at a pitch angle θ with respect to the magnetic field $\mathbf{B}$:

Uniform static
magnetic field



$$\frac{d}{dt}(\gamma m_0 \mathbf{v}) = ze(\mathbf{v} \times \mathbf{B})$$
$$\Rightarrow r = \frac{\gamma m_0 |\mathbf{v}| \sin \theta}{ze |\mathbf{B}|}$$

- Particle's acceleration vector is perpendicular to the magnetic field direction $\mathbf{B}$ and to $\mathbf{v}_\perp$
- Motion consists of a constant velocity along the direction of $\mathbf{B}$ and circular motion about it with radius r .
- **Particle moves in a spiral path with constant radius and pitch angle.**

Acceleration of high-energy particles

Dynamics of charged particles in a uniform static magnetic field

We can rewrite the expression for the radius of the particle's path as:

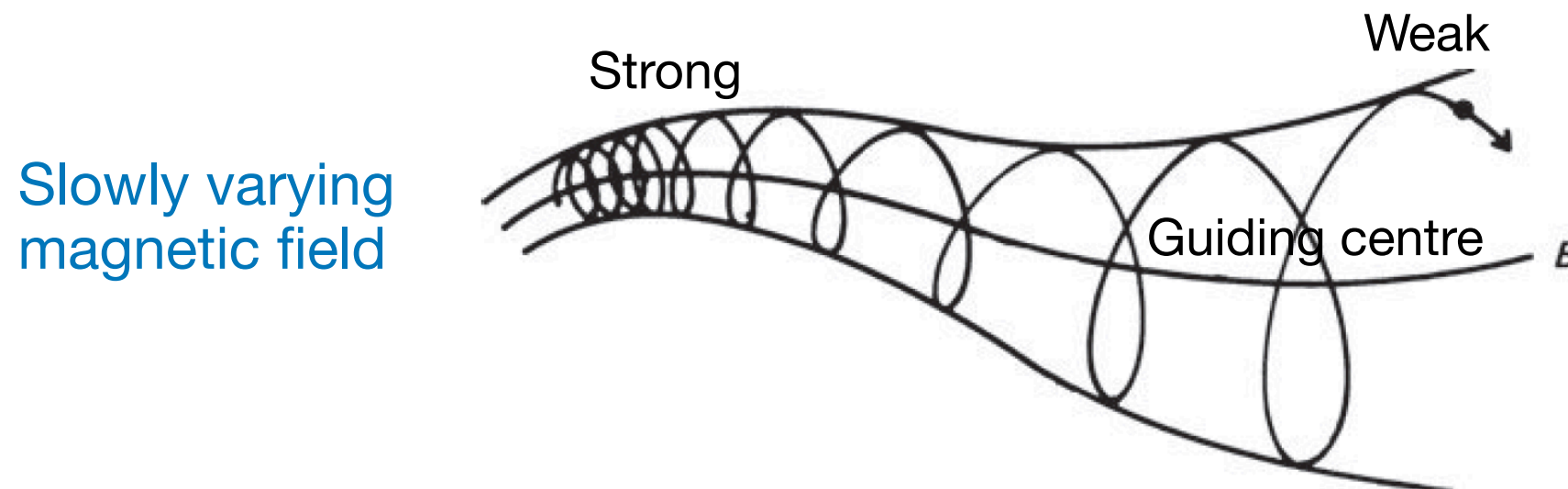
$$r = \frac{\gamma m_0 v}{ze} \frac{\sin \theta}{|\mathbf{B}|} = \left(\frac{pc}{ze} \right) \frac{\sin \theta}{|\mathbf{B}|c}$$

where $p = \gamma m_0 |\mathbf{v}|$ is the relativistic three-momentum of the particle.

- Particles with the same values of pc/ze and pitch angle θ into a magnetic field $\mathbf{B}$ behave exactly the same way.
- The quantity pc/ze is called the rigidity or magnetic rigidity of the particle. Its unit/dimension is energy/charge. For example: $GeV/e \rightarrow GV$.

Acceleration of high-energy particles

Dynamics of charged particles in a slowly varying magnetic field



- Particle's guiding centre follows the mean magnetic field direction.
- **Particle follows the guiding centre in such a way that the number of field lines within the particle's orbit remain constant.**
- Changing magnetic field in one orbit induce electric field which accelerate the particle in the radial direction, leading to a change in the kinetic energy in the radial direction.
- Total kinetic energy of the particle remains constant. So, the change in the radial kinetic energy comes from the change in the parallel kinetic energy.
- **Particle moves in a spiral path with changing radius and a slow change in pitch angle.**

Acceleration of high-energy particles

Scattering of charged particles by irregularities in the magnetic field

- Charged particles follow the mean magnetic field direction for a slowly varying field.
- Irregularities are present in the magnetic field which can scatter the particle (a large pitch angle change in one orbit).
- If these scatterings are random, the result is a uniform distribution of pitch angles
=> Diffusive motion of charged particles.
- Example of irregularities: Presence of magnetic fluctuations in the interplanetary magnetic field measured by the Mariner 4 space probe on its way to Mars.

But, not all fluctuations can scatter the particles!

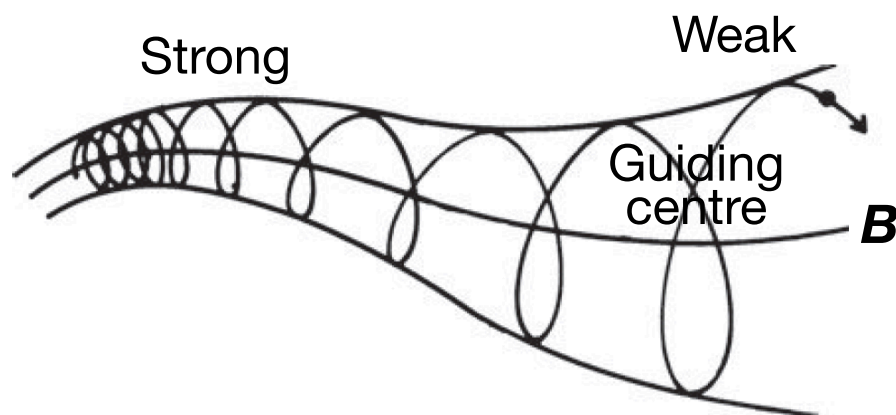
- Scattering depends on the relative size of the particle gyroradius and the scale of the fluctuation
- Wave-particle resonant interaction ($r_g \sim \lambda$).

Acceleration of high-energy particles

Scattering of charged particles by irregularities in the magnetic field

(a)

Gyroradius $\ll$ Fluctuation scale



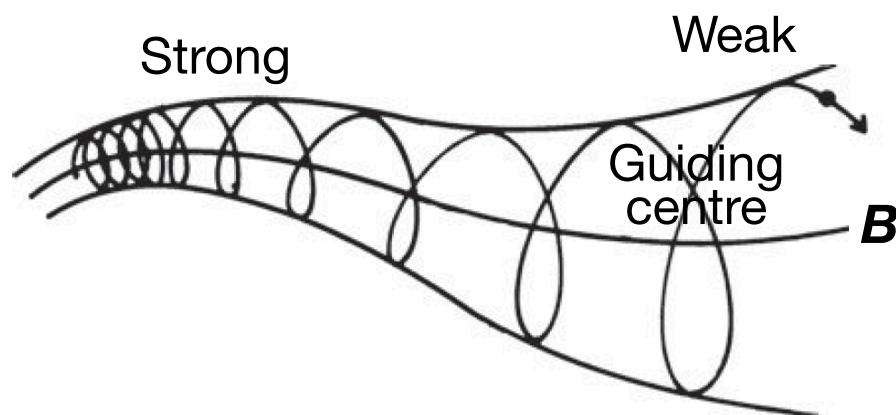
- Particle follows guiding centre.
- Small change in pitch angle.

Acceleration of high-energy particles

Scattering of charged particles by irregularities in the magnetic field

(a)

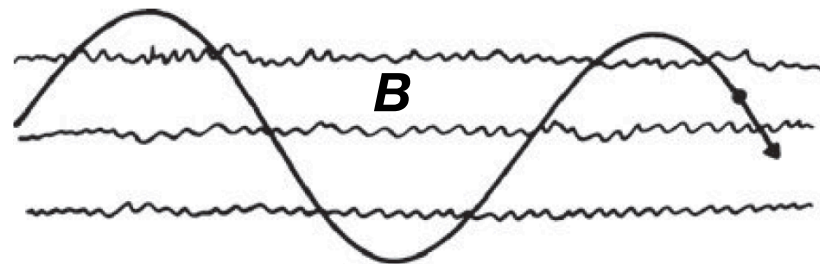
Gyroradius $\ll$ Fluctuation scale



- Particle follows guiding centre.
- Small change in pitch angle.

(b)

Gyroradius $\gg$ Fluctuation scale



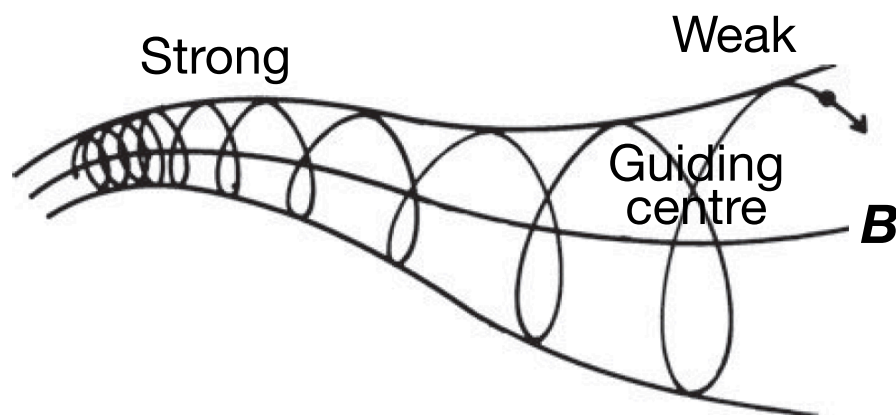
- Particle do not feel the fine structure of the field.
- Move in orbits determined by the mean magnetic field.
- Almost no change in pitch angle.

Acceleration of high-energy particles

Scattering of charged particles by irregularities in the magnetic field

(a)

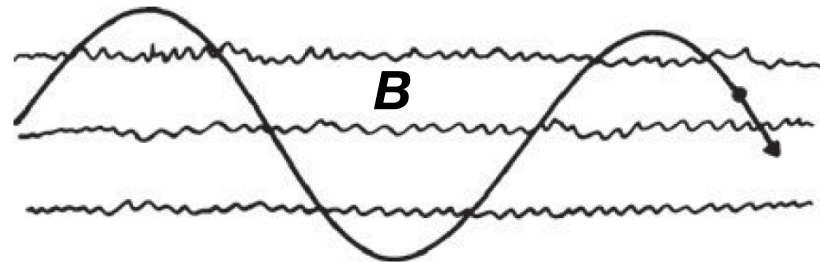
Gyroradius $\ll$ Fluctuation scale



- Particle follows guiding centre.
- Small change in pitch angle.

(b)

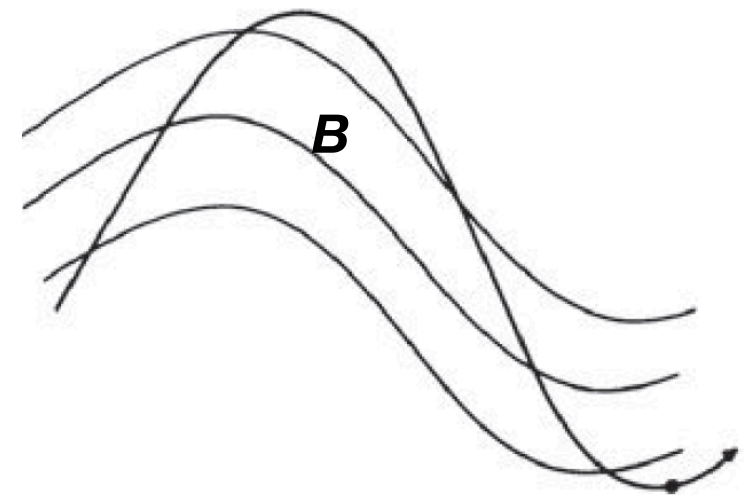
Gyroradius $\gg$ Fluctuation scale



- Particle do not feel the fine structure of the field.
- Move in orbits determined by the mean magnetic field.
- Almost no change in pitch angle.

(c)

Gyroradius $\sim$ Fluctuation scale



- Only in this case, significant change in pitch angle in one gyroradius happens.
- Scattering by the random superposition of such fluctuations leads to stochastic changes in the pitch angle.

Acceleration of high-energy particles

Scattering of charged particles by irregularities in the magnetic field

- In most astrophysical environment, magnetic field irregularities are superimposed randomly.
- Particle 'feels' the influence of a particular field irregularity for about one wavelength before it encounters another irregularity with random phase relative to the last wave.
- The result is a large and random change in the pitch angle of the particle in a single wavelength.
- Over a large distance, the motion of the particle becomes diffusive.

Random pitch angle scattering of particle => diffusive motion

Acceleration of high-energy particles

Scattering of charged particles by Alfvén and hydromagnetic waves

- Alfvén waves are transverse magnetic waves that can be excited in a partially or fully ionized plasma embedded in a magnetic field. They travel along magnetic field lines.

Acceleration of high-energy particles

Scattering of charged particles by Alfvén and hydromagnetic waves

- Alfvén waves are transverse magnetic waves that can be excited in a partially or fully ionized plasma embedded in a magnetic field. They travel along magnetic field lines.
- In astrophysical environments, these waves can be generated/excited by the streaming of high-energy charged particles.
- These waves generate magnetic fluctuations that can scatter charged particles.
- Scattering is wave-particle resonant interaction ($r_g \sim \lambda$).

Acceleration of high-energy particles

Scattering of charged particles by Alfvén and hydromagnetic waves

- Alfvén waves are transverse magnetic waves that can be excited in a partially or fully ionized plasma embedded in a magnetic field. They travel along magnetic field lines.
- In astrophysical environments, these waves can be generated/excited by the streaming of high-energy charged particles.
- These waves generate magnetic fluctuations that can scatter charged particles.
- Scattering is wave-particle resonant interaction ($r_g \sim \lambda$).
- The amplitude of the waves grow as forward momentum of the streaming particles is transferred to the waves.
- The growth continues until the streaming velocity of the high-energy particles is reduced to the Alfvén velocity $v_A = B_0/(\mu_0\rho)^{1/2}$

where B_0 is the mean magnetic flux,

μ_0 is the permeability of the vacuum,

$\rho = N_p m_p$ is the mass density of the ionized component of the plasma medium.

- For the interstellar medium, taking $N_p \sim 0.1 \text{ cm}^{-3}$ and $B_0 = 3 \text{ } \mu\text{G}$, $v_A \sim 20 \text{ km/s}$.

Acceleration of high-energy particles

Shock waves

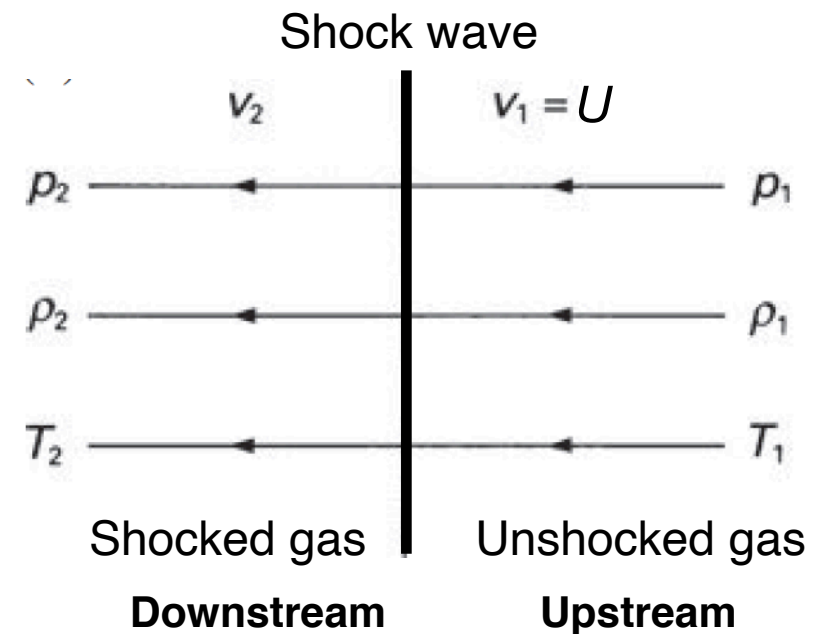
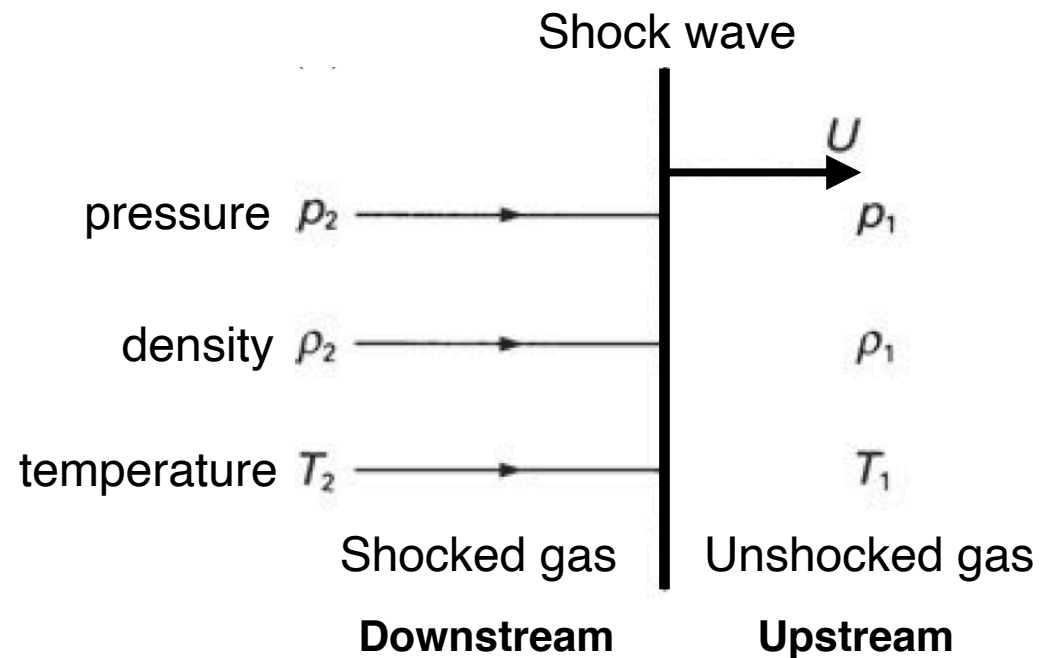
- Perturbations are propagated away from their source at the speed of sound in the medium.
- If a disturbance is propagated at a velocity greater than the speed of sound, it cannot behave like a sound wave.
- There is a discontinuity between the regions behind and ahead of the disturbance.
- These discontinuities are called shock waves.

Acceleration of high-energy particles

Shock waves

A shock wave propagating through a stationary gas at a supersonic velocity U

In the rest frame of the shock



The behaviour of the gas on passing through the shock wave is described by a set of conservation relations:

- (1) Mass conservation on passing through the discontinuity.
- (2) Continuity of energy flux.
- (3) Continuity of momentum flux.

Acceleration of high-energy particles

Shock waves

In the limit of very strong shocks, $M_1 \gg 1$,

$$\begin{aligned}\frac{p_2}{p_1} &= \frac{2\gamma M_1^2}{(\gamma + 1)}, \\ \frac{\rho_2}{\rho_1} &= \frac{v_1}{v_2} = \frac{(\gamma + 1)}{(\gamma - 1)}, \\ \frac{T_2}{T_1} &= \frac{2\gamma(\gamma - 1)M_1^2}{(\gamma + 1)^2}.\end{aligned}$$

where,

the Mach number $M_1 = U/c_1 = v_1/c_1$

$c_1 = (\gamma p_1/\rho_1)^{1/2}$ is the velocity of sound in the unshocked gas (upstream)

γ is the ratio of specific heat capacities

Acceleration of high-energy particles

Shock waves

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$c_1 = (\gamma p_1/\rho_1)^{1/2}$ is the velocity of sound in the unshocked gas (upstream)

γ is the ratio of specific heat capacities

For a monoatomic gas, $\gamma = \frac{5}{3}$. This gives $\frac{\rho_2}{\rho_1} = \frac{v_1}{v_2} = 4$ Also, $\frac{B_2}{B_1} = 4$

For $v_1=U$, this implies $v_2=(1/4)U$. So, $v_1-v_2=(3/4)U$.

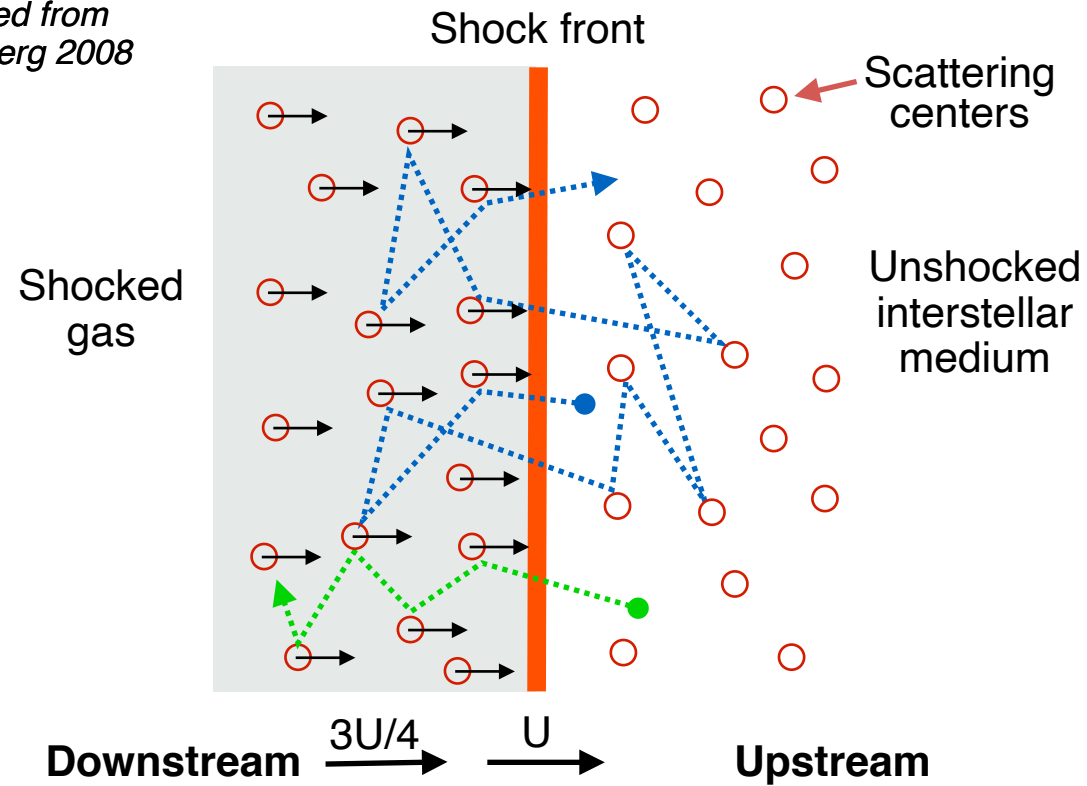
$\frac{v_1}{v_2} = r$ is referred to as the shock compression ratio. $r = 4$ for strong shocks.

Diffusive shock acceleration of charged particles

First order Fermi acceleration (The transport equation approach)

Interstellar rest frame

Adapted from
Achterberg 2008

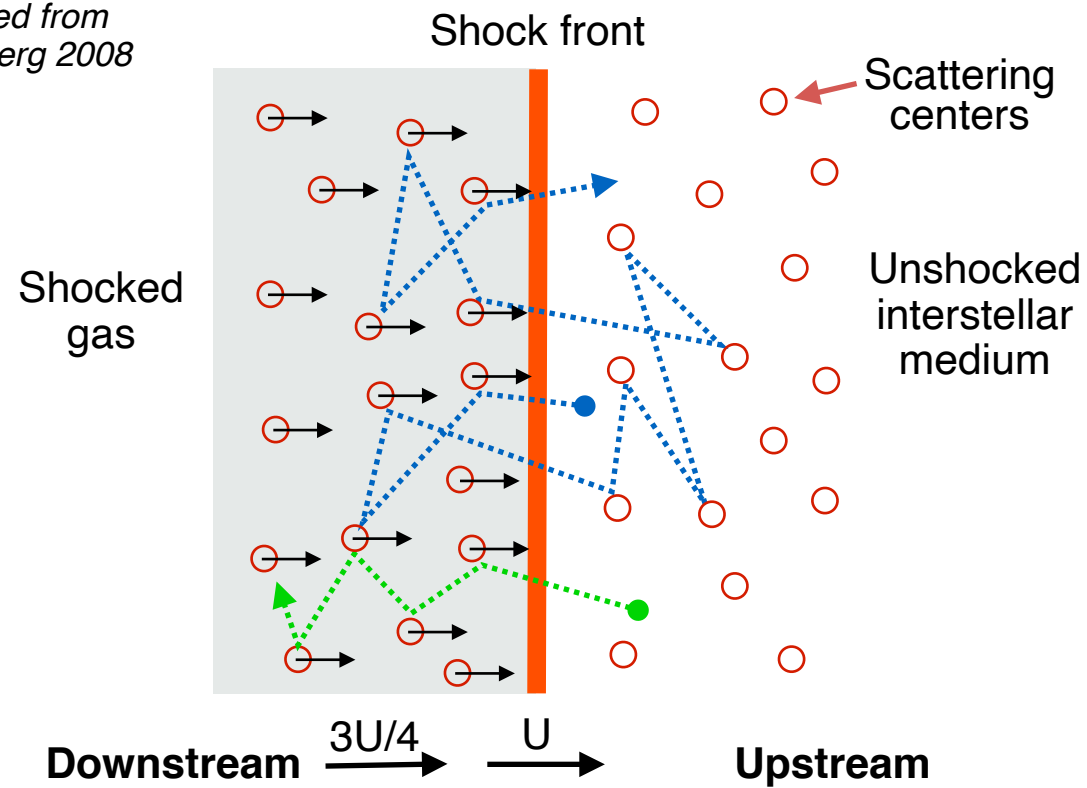


Diffusive shock acceleration of charged particles

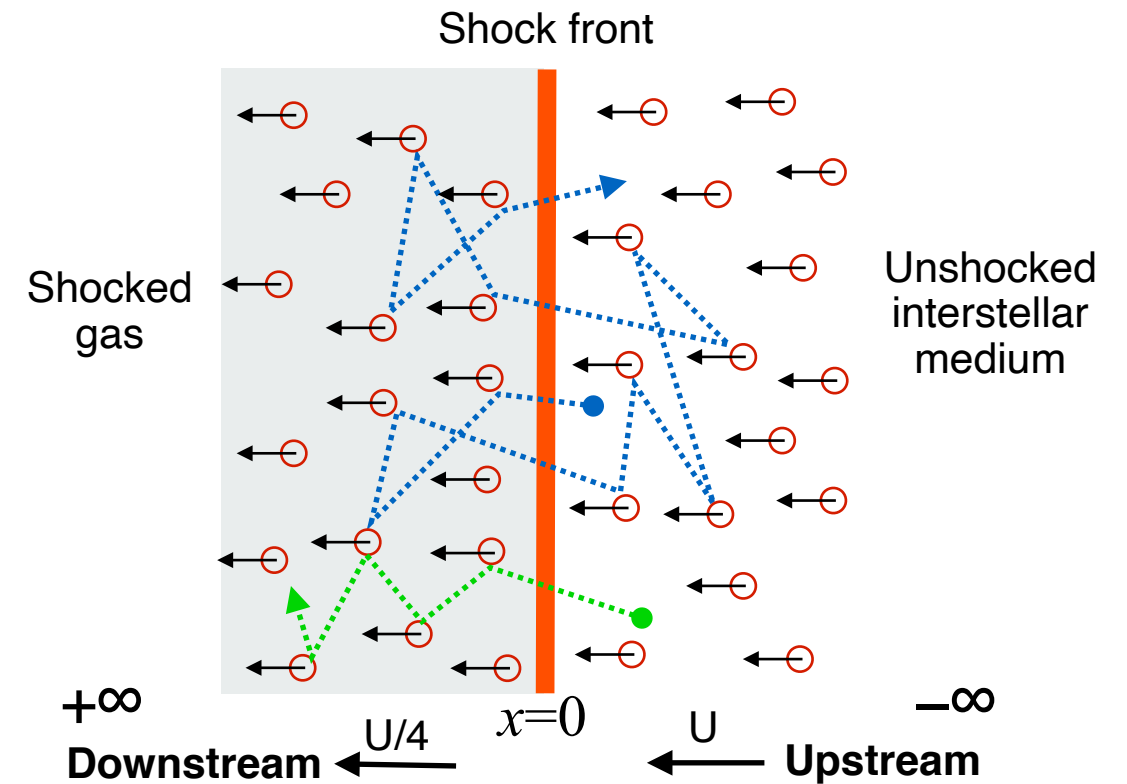
First order Fermi acceleration (The transport equation approach)

Interstellar rest frame

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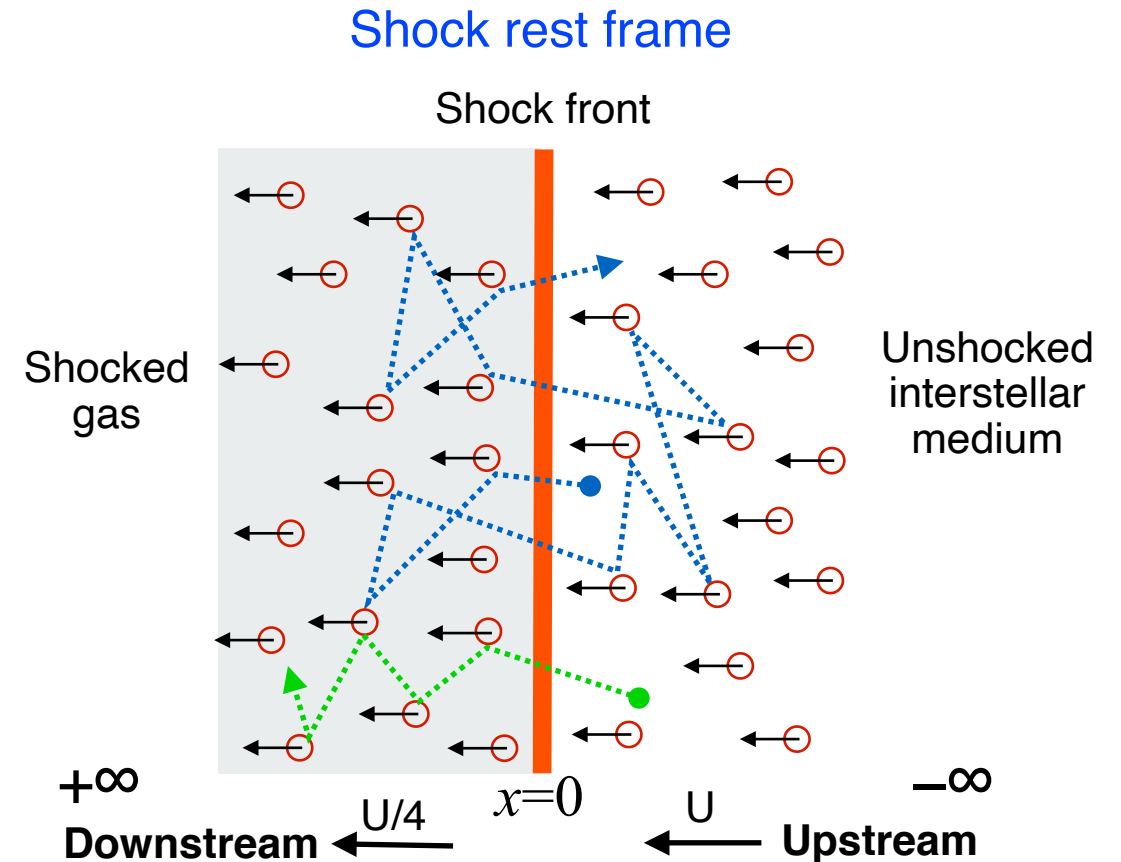
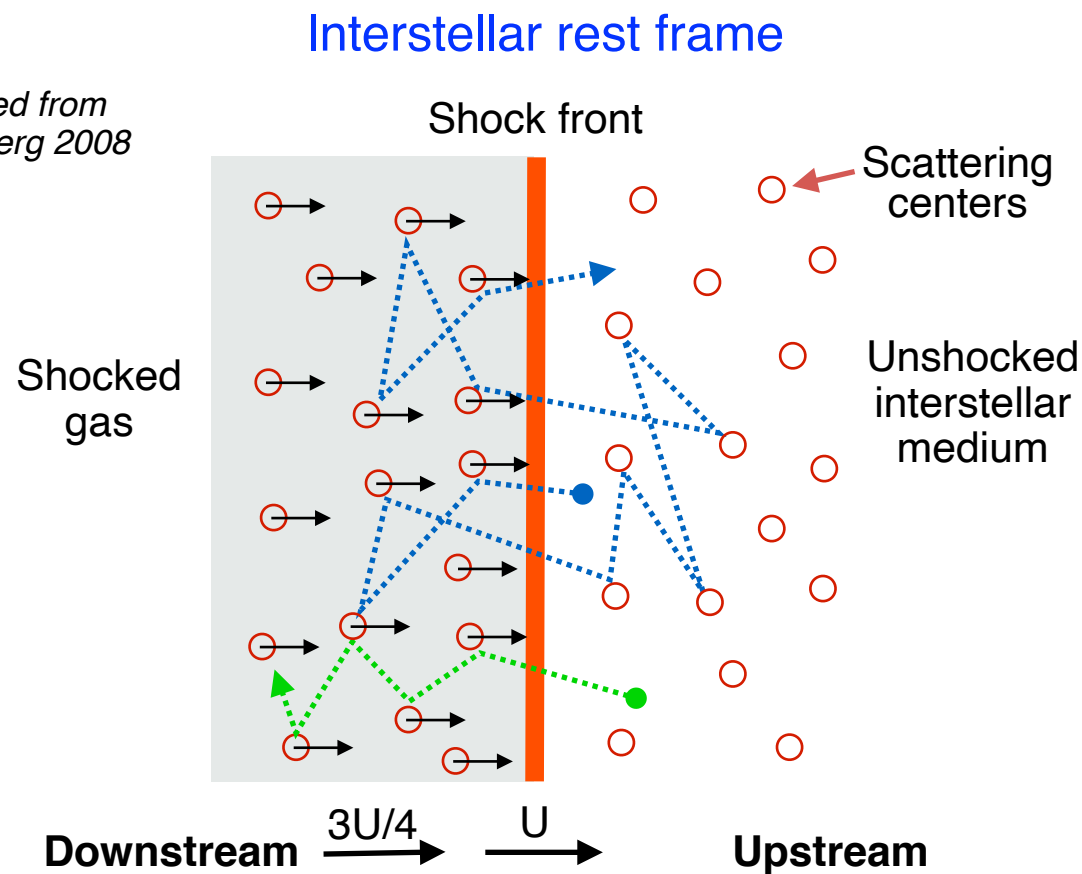
Shock rest frame



Diffusive shock acceleration of charged particles

First order Fermi acceleration (The transport equation approach)

Adapted from
Achterberg 2008



Transport equation for particle acceleration in the shock rest frame

$$\frac{\partial}{\partial x} \left(D \frac{\partial f}{\partial x} \right) - u \frac{\partial f}{\partial x} + \frac{1}{3} \frac{du}{dx} p \frac{\partial f}{\partial p} + Q(x, p) = 0$$

Diffusion Advection Compression Injection

Note the
steady-state
equation

$f(p)$: Particle distribution function (phase space density) $\rightarrow \# / (\text{volume} \times \text{momentum}^3 \times \text{energy})$

The particle density $N(p)$ is related to $f(p)$ as: $N(x, p) dp = 4\pi p^2 f(x, p) dp$

$D \rightarrow$ Diffusion coefficient. Can be assumed to be same on both sides of the shock for simplicity.

Diffusive shock acceleration of charged particles

The particle spectrum from diffusive shock acceleration

$$\frac{\partial}{\partial x} \left(D \frac{\partial f}{\partial x} \right) - u \frac{\partial f}{\partial x} + \frac{1}{3} \frac{du}{dx} p \frac{\partial f}{\partial p} + Q(x, p) = 0$$

Diffusion Advection Compression Injection

Transport equation
in the shock rest frame

The original equation in 3D:

$$\nabla \cdot (D \nabla N - \mathbf{u} N) + \frac{\partial}{\partial p} \left(\frac{\nabla \cdot \mathbf{u}}{3} p N \right) + q(x, p) = 0 \quad (1)$$

$D(p)$: Diffusion coefficient (can be assumed to be spatial independent)

$N(x, p)$: Number density of the particles

$\mathbf{u}$: velocity of the gas

p : Momentum of the high-energy particle

$q(x, p)$: Rate of injection of the thermal particles $1/(cm^3 \text{ sec})$.

Diffusive shock acceleration of charged particles

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Considering the problem in 1D (along x-axis): $\mathbf{u} = u\hat{u}$ and writing $q(x, p)dp = 4\pi p^2 Q(x, p)dp$

Eq. (1) reduces to:
$$\frac{\partial}{\partial x} \left(D \frac{\partial f}{\partial x} - u f \right) + \frac{1}{3} \frac{\partial u}{\partial x} \left(3f + p \frac{\partial f}{\partial p} \right) + Q(x, p) = 0 \quad (2)$$

Rearranging:
$$\boxed{\frac{\partial}{\partial x} \left(D \frac{\partial f}{\partial x} \right) - u \frac{\partial f}{\partial x} + \frac{1}{3} \frac{du}{dx} p \frac{\partial f}{\partial p} + Q(x, p) = 0} \quad (3)$$

Diffusive shock acceleration of charged particles

The particle spectrum from diffusive shock acceleration

$$\boxed{\frac{\partial}{\partial x} \left(D \frac{\partial f}{\partial x} \right) - u \frac{\partial f}{\partial x} + \frac{1}{3} \frac{du}{dx} p \frac{\partial f}{\partial p} + Q(x, p) = 0}$$

Diffusion Advection Compression Injection

Transport equation
in the shock rest frame

- * Solve this equation for $f(p)$ using the boundary conditions: $f(p) = 0$ at $x = -\infty$ (Upstream)
 $f(p) = \text{finite}$ at $x = +\infty$ (Downstream)

- * Also, the velocity jump condition $\frac{du}{dx} = (u_2 - u_1)\delta(x)$

where, $u_1(u_2)$ upstream (downstream) gas velocities with respect to the shock.

Note that $u_1 = U$ is the velocity of the shock in the gas medium.

Diffusive shock acceleration of charged particles

The particle spectrum from diffusive shock acceleration

$$\boxed{\frac{\partial}{\partial x} \left(D \frac{\partial f}{\partial x} \right) - u \frac{\partial f}{\partial x} + \frac{1}{3} \frac{du}{dx} p \frac{\partial f}{\partial p} + Q(x, p) = 0}$$

Diffusion Advection Compression Injection

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Note that $u_1 = U$ is the velocity of the shock in the gas medium.

- * The injection of thermal “seed” particles having momentum p_i happens at the shock front ($x=0$).
So, the injection term can be written as $Q(x, p) = q_0(p)\delta(x)$ where $q_0(p) = \frac{n_i}{4\pi p_i^2} u_1 \delta(p - p_i)$

Diffusive shock acceleration of charged particles

The particle spectrum from diffusive shock acceleration

$$\boxed{\frac{\partial}{\partial x} \left(D \frac{\partial f}{\partial x} \right) - u \frac{\partial f}{\partial x} + \frac{1}{3} \frac{du}{dx} p \frac{\partial f}{\partial p} + Q(x, p) = 0}$$

Diffusion Advection Compression Injection

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[Hint: Solve for the downstream and upstream regions separately, and connect the solutions at $x = 0$ using the continuity of the solution at $x=0$. The continuity equation can be obtained by integrating Eq (2) on slide 38 around $x = 0$ (from $x = 0^-$ to $x = 0^+$)]

Diffusive shock acceleration of charged particles

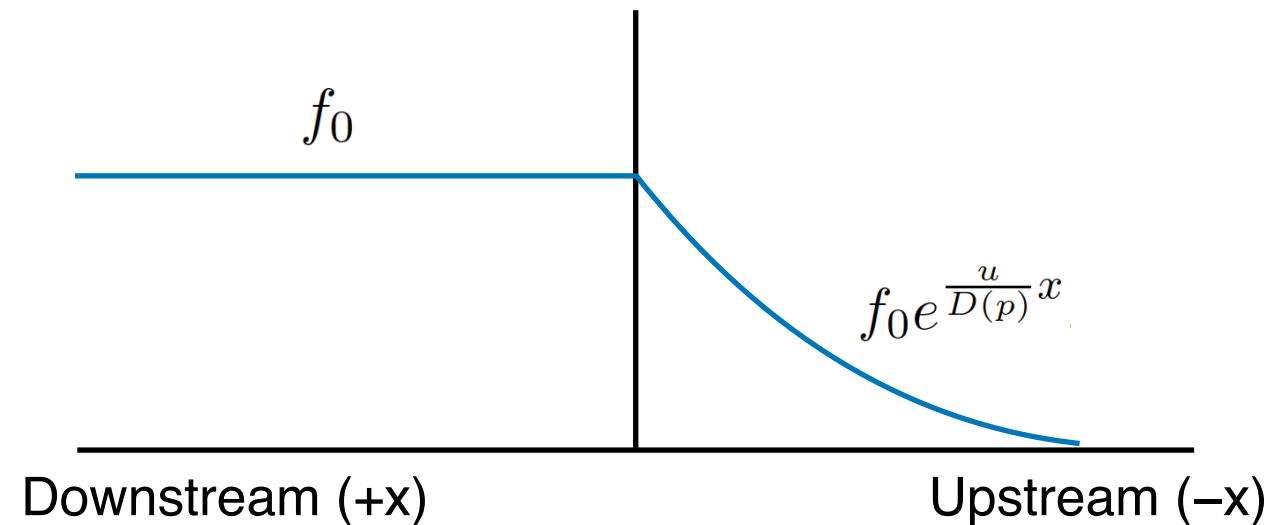
***Upstream/downstream equations:** $\frac{\partial}{\partial x} \left(D \frac{\partial f}{\partial x} \right) - u \frac{\partial f}{\partial x} = 0$

Solution: $f(p) = f_0$, downstream ($x > 0$)

$f(p) = f_0 e^{\frac{u_1}{D(p)} x}$, upstream ($x < 0$)

where, $f_0 = \frac{\gamma n_i}{4\pi p_i^3} \left(\frac{p}{p_i} \right)^{-\gamma}$

$$\gamma = \frac{3u_1}{u_1 - u_2} = \frac{3r}{r - 1}$$



Note: This will be the question for the next assignment.

Diffusive shock acceleration of charged particles

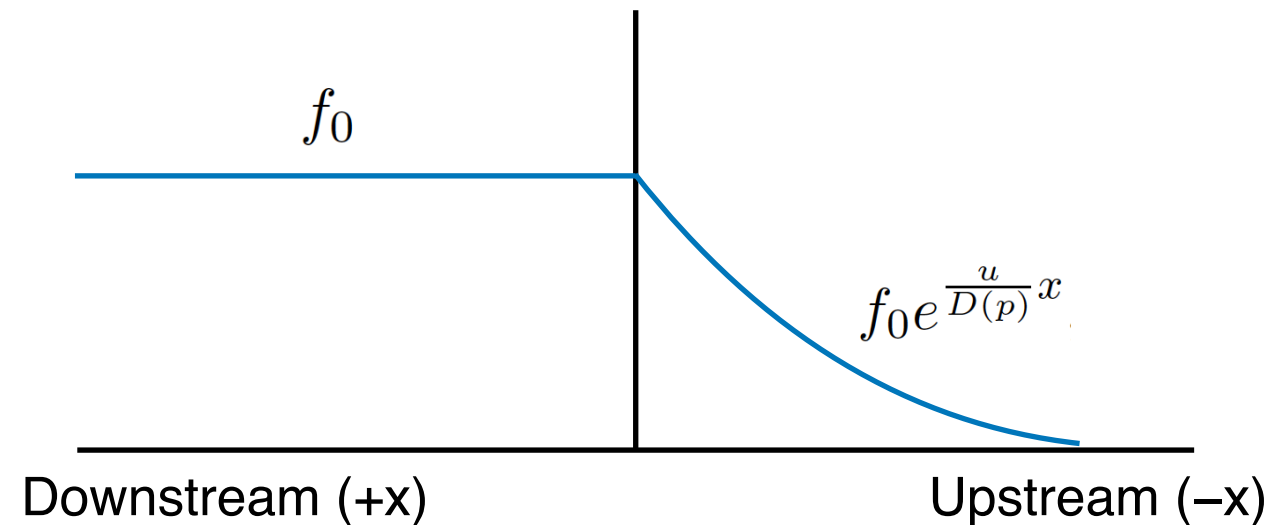
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$$\gamma = \frac{3u_1}{u_1 - u_2} = \frac{3r}{r - 1}$$



*For strong shocks, the compression ratio $r = 4$. This gives $\gamma = 4$

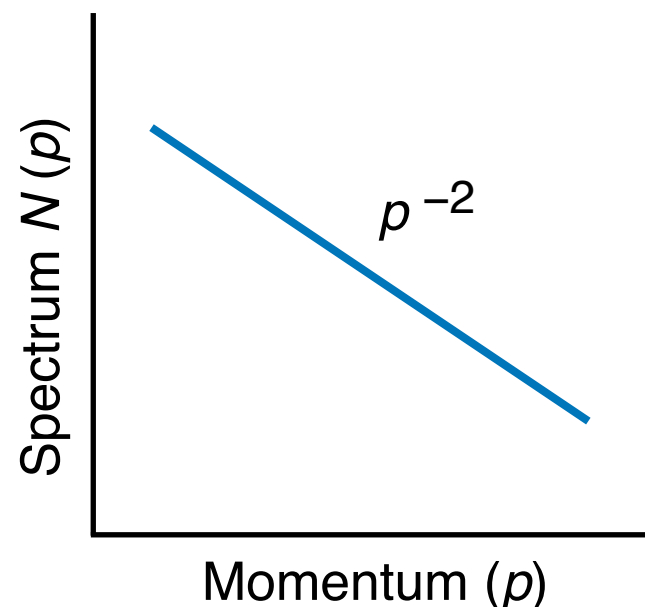
*So, we obtain $N(p)dp = 4\pi p^2 f(p)dp \Rightarrow N(p) \propto p^{-2}$

The spectrum of the accelerated particles follows a power law of index 2!

Diffusive shock acceleration of charged particles

Some important conclusions

- * The spectrum does not depend on the mechanism causing the scattering as long as the particles are bouncing across the shock.
- * The spectral index only depends on the compression ratio (u_1/u_2). For strong shocks found in a variety of astrophysical objects, the index is universal and close to 2.
- * The steady-state calculation cannot estimate the maximum energy of the particles.



Diffusive shock acceleration of charged particles

Corresponding equation for electrons:

$$\nabla \cdot (D \nabla N - \mathbf{u} N) + \frac{\partial}{\partial p} \left[\left(\frac{\nabla \cdot \mathbf{u}}{3} p + B(p) \right) N \right] + q(x, p) = 0$$

$D(p)$: Diffusion coefficient (can be assumed to be spatial independent)

$N(x, p)$: Number density of the particles

$\mathbf{u}$: velocity of the gas

p : Momentum of the high-energy particle

$B(p) \propto p^2$ is the energy loss rate mainly due to synchrotron/inverse Compton

$q(x, p)$: Rate of injection of the thermal particles $1/(cm^3 \text{ sec})$.

In 1D, the equation becomes:

$$\frac{\partial}{\partial x} \left(D \frac{\partial N}{\partial x} - u N \right) + \frac{\partial}{\partial p} \left[\left(\frac{1}{3} \frac{\partial u}{\partial x} p + B(p) \right) N \right] + Q(x, p) = 0$$

No exact analytical solution obtained so far!

Diffusive shock acceleration of charged particles

Maximum energy of the accelerated particles

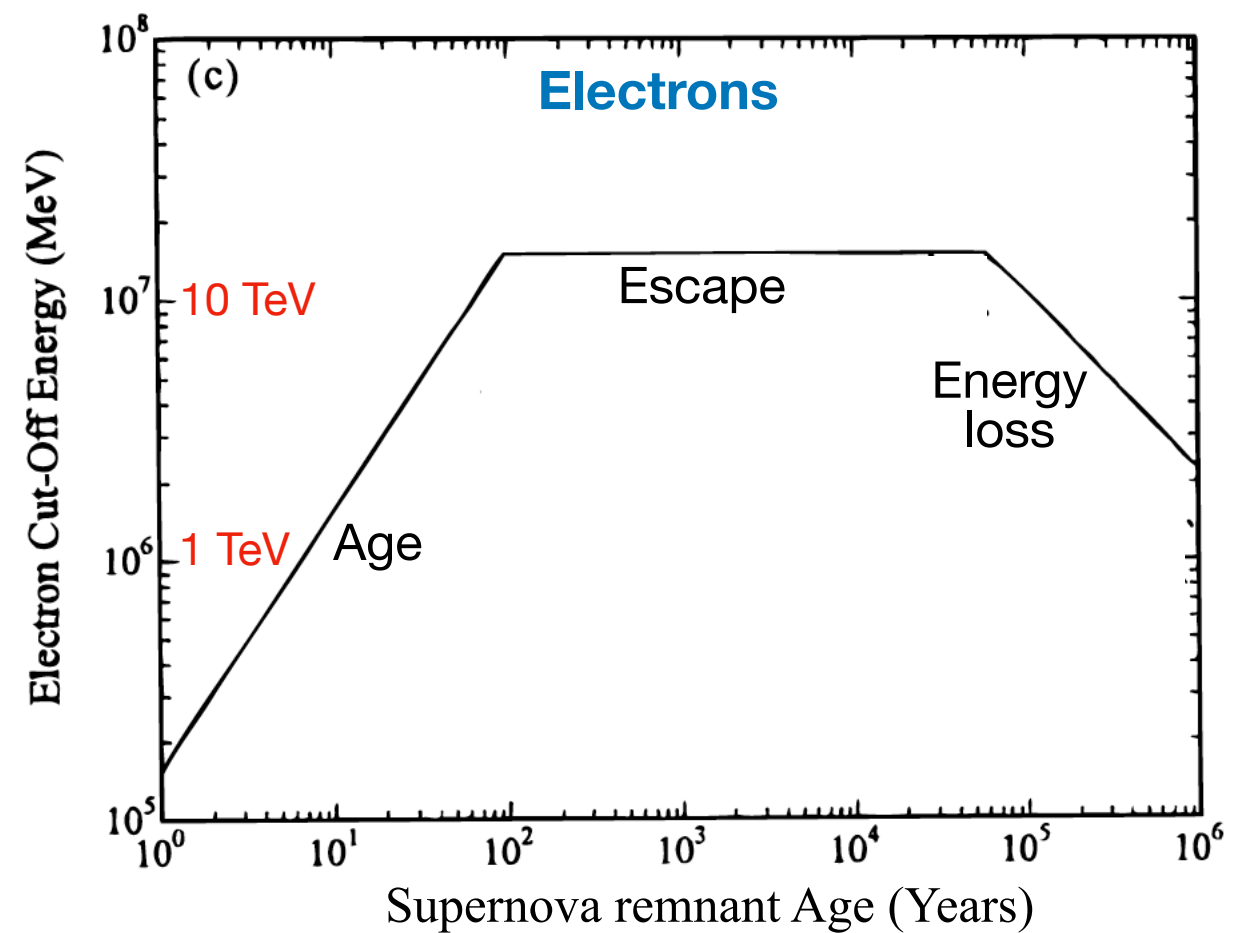
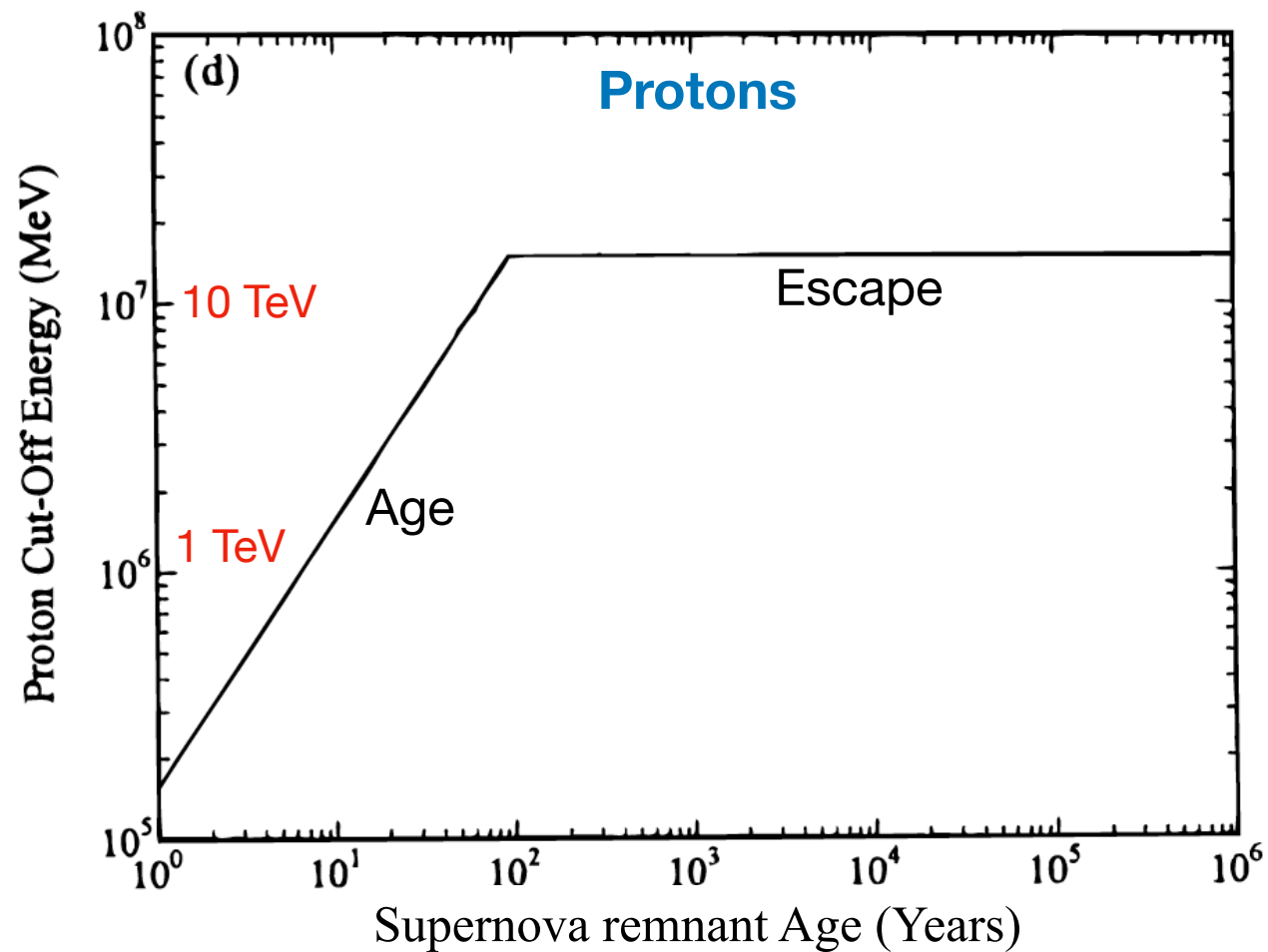
Maximum energy is determined by three different mechanisms:

- (1) Finite age of the shock,
- (2) Free escape from the shock region when the particles are no longer effectively scattered by the magnetic irregularities,
- (3) Energy-loss processes such as synchrotron or inverse Compton (relevant for electrons).

The minimum time of (1), (2) and (3) decides the maximum energy.

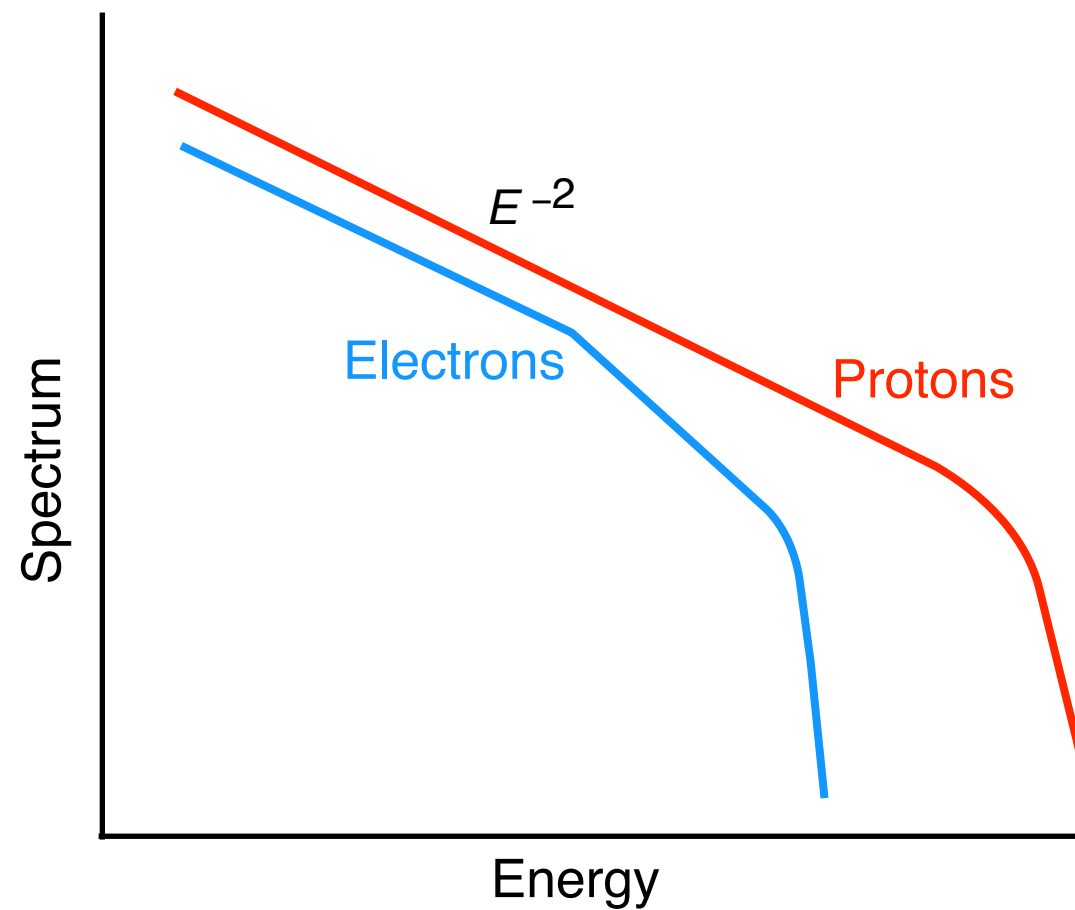
Diffusive shock acceleration of charged particles

Maximum energy of the accelerated particles



Diffusive shock acceleration of charged particles

Expected spectrum of particles inside a supernova remnant after $\sim 10^5$ years



**Next, we will learn about the evidence of particle acceleration
in Astrophysical objects.**